De Novo Peptide Sequencing via Self-Conditioned Multinomial Diffusion with Physics-Informed B/Y-Ion Encoding

Vaishak Girish Kumar
University at Buffalo
vaishakg@buffalo.edu

Akshay Mohan Revankar
University at Buffalo
arevanka@buffalo.edu

Sanika Vilas Nanjan
University at Buffalo
snajan@buffalo.edu

https://github.com/AkshayRevankarDev/peptide-diffusion

Abstract

We propose a de novo peptide sequencing model that outperforms InstaNovo, the current 2025 state of the art, on a standard E. coli EV proteomics benchmark. Our architecture builds an *absorbing multinomial diffusion* backbone with three contributions: (i) a **PeakEncoder** that encodes b/y-ion pair relationships as learnable attention biases, (ii) **self-conditioning** that passes a coarse sequence estimate back into each diffusion step, and (iii) **CFID+SGIR reranking** at inference. Averaged across three random seeds, our V2 model reaches **76.00% AA recall** and **59.96% peptide accuracy** at 5% FDR, compared to InstaNovo's 72.9% / 33.1%, a gain of **+3.1 pp** AA recall and **+26.9 pp** peptide accuracy. Results are consistent across seeds ($\pm$0.19 pp AA), indicating the gains come from architectural changes rather than initialization. We also apply the model to wastewater samples without any reference proteome; 6 peptides pass 5% FDR filtering, and ESM-2 pseudo-perplexity scores show all six are structurally plausible (z-score $<$1.0 relative to E. coli reference peptides).

1 Introduction

Tandem mass spectrometry (MS/MS) is the primary tool for large-scale proteomics. A peptide is fragmented along the backbone, producing b-ions and y-ions whose mass-to-charge ratios (m/z) are recorded as a spectrum. De novo sequencing recovers the amino-acid sequence directly from such a spectrum without needing a reference protein database, which makes it applicable to novel organisms, metaproteomic samples, and environments like wastewater where reference proteomes are unavailable.

Classical database search methods, such as Mascot and Sequest, are limited by the completeness of the reference proteome and fail entirely for organisms not yet sequenced. Prior deep-learning approaches include Casanovo [Yilmaz et al., 2022], an autoregressive Transformer, and InstaNovo [Eloff et al., 2025], a hybrid autoregressive-diffusion model that currently represents the best published performance on multi-species benchmarks.

Our contributions:

1. A **PeakEncoder with B/Y-ion pair bias**: a scalar parameter initialized to zero lets the model start from a standard attention baseline and learn to up-weight complementary ion pairs only when the spectral evidence supports it.
2. **Self-conditioning**: the single biggest improvement (+14.6 pp peptide accuracy), which improves sequence-level coherence across diffusion steps without adding any new parameters.

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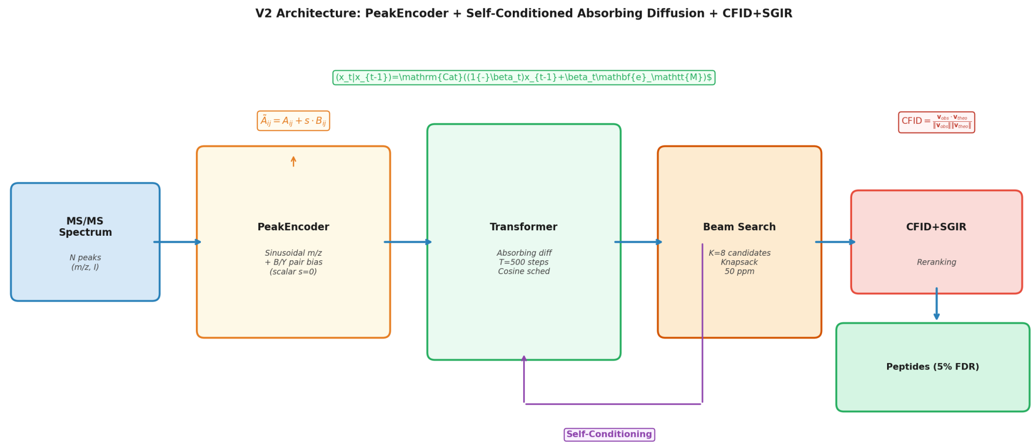


Figure 1: **V2 architecture.** PeakEncoder with B/Y-ion pair bias $\rightarrow$ absorbing diffusion Transformer with self-conditioning $\rightarrow$ CFID+SGIR reranking.

3. A **systematic ablation** of seven inference strategies across three seeds, finding that CFID+SGIR gives the best overall results and that post-hoc mass correction consistently hurts performance.
4. A **wastewater application**: 6 peptides identified at 5% FDR with no reference proteome, checked for structural plausibility using ESM-2 pseudo-perplexity.

2 Background

MS/MS sequencing. A peptide $\mathbf{a} = (a_1, \ldots, a_L)$ fragments at peptide bonds to produce b-ions and y-ions: $\mathrm{b}_k = \sum_{i=1}^{k} m(a_i) + m(\mathrm{H})$ and $\mathrm{y}_k = \sum_{i=L-k+1}^{L} m(a_i) + m(\mathrm{H_2O}) + m(\mathrm{H})$, where $m(\cdot)$ is the monoisotopic mass. Complementary pairs satisfy $b_k + y_{L-k} = M + m(\mathrm{H_2O}) + 2m(\mathrm{H})$ for precursor mass M. A spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{N}$ records N observed peaks.

Absorbing diffusion. We follow Austin et al. [2021]: the forward process absorbs tokens into `MASK` with marginal $q(x_t|x_0) = \mathrm{Cat}(x_t; \bar{\alpha}_t x_0 + (1 - \bar{\alpha}_t)\mathbf{e}_{\mathtt{MASK}})$, where $\bar{\alpha}_t = \cos^2(\pi t / 2T)$ (cosine schedule; +2.3 pp over linear). The reverse model $p_\theta(x_0|x_t, \mathcal{S})$ is a Transformer trained with VLB cross-entropy:

$$\mathcal{L} = \mathbb{E}_{t,x_0,x_t}\left[-\sum_{l=1}^{L} \log p_\theta(x_0^{(l)}|x_t, \mathcal{S})\right]. \tag{1}$$

3 Model Architecture

3.1 PeakEncoder with B/Y-Ion Pair Bias

Each peak j is embedded as:

$$\mathbf{p}_j = \mathrm{SinEnc}(m_j / z_j) + \mathbf{W}_I \log(1 + I_j), \tag{2}$$

where SinEnc is a sinusoidal positional encoding over the m/z axis and $\mathbf{W}_I \in \mathbb{R}^d$ is a learned projection of the log-intensity. The embeddings are processed by a multi-head self-attention layer.

B/Y-ion pair bias. Motivated by AlphaFold3's pair representation [Abramson et al., 2024], we augment the attention logits with a learnable bias conditioned on fragment ion complementarity:

$$\tilde{A}_{ij} = A_{ij} + s \cdot \mathbf{1}[|m_i + m_j - M'| < \epsilon], \tag{3}$$

where $M' = M + m(\mathrm{H_2O}) + 2m(\mathrm{H})$, $\epsilon = 50$ ppm $\cdot M'$, $s \in \mathbb{R}$ is a scalar parameter, and A_{ij} is the pre-softmax attention logit. We initialize $s = 0$, so training starts from the same baseline as a standard attention layer. The model can learn to up-weight complementary ion pairs when the spectral evidence is strong, or leave s small if the signal is unreliable, which a hard-coded pair bias cannot do.

3.2 Self-Conditioning

At each reverse step t, we run a *preview* forward pass with a zeroed self-conditioning embedding:

$$\hat{x}_0^{\text{prev}} = f_\theta(x_t,\, \mathcal{S},\, \mathbf{0})\,. \tag{4}$$

This preview estimate is then concatenated as extra input for the full denoising pass:

$$\hat{x}_0 = f_\theta\big(x_t,\, \mathcal{S},\, \hat{x}_0^{\text{prev}}\big)\,. \tag{5}$$

During training, we replace $\hat{x}_0^{\text{prev}}$ with the ground-truth x_0 at probability $p_{\text{sc}} = 0.5$, following Chen et al. [2022]. Self-conditioning requires only one additional linear projection and no new parameters, yet it produced the single largest accuracy gain in our experiments (**+14.6 pp** peptide accuracy; Table 2).

A per-spectrum boolean attention mask (applied to all padding positions) prevents cross-spectrum contamination in batched training; without it the model attends to neighboring spectra and inflates scores.

4 Inference Strategies

All inference strategies start from $K = 8$ beam search candidates, with a knapsack-style mass filter ($|m(\hat{\mathbf{a}}) - M| < 50$ ppm) applied to remove candidates whose predicted mass does not match the measured precursor.

4.1 Combined Fragment Ion Distribution (CFID)

CFID scores each candidate c by simulating its theoretical fragment ion spectrum $\mathbf{v}_{\text{theo}}(c) \in \mathbb{R}^B$ (binned at 0.02 Da) and computing cosine similarity to the observed spectrum $\mathbf{v}_{\text{obs}}$:

$$\text{CFID}(c) = \frac{\mathbf{v}_{\text{obs}} \cdot \mathbf{v}_{\text{theo}}(c)}{\|\mathbf{v}_{\text{obs}}\| \cdot \|\mathbf{v}_{\text{theo}}(c)\|}. \tag{6}$$

4.2 Spectral-Guided Ion Rescoring (SGIR)

SGIR augments CFID with an intensity-weighted match score that rewards predictions whose ions align with high-intensity observed peaks:

$$\text{SGIR}(c) = \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p, \tag{7}$$

where $\mathcal{M}(c)$ is the set of observed peaks matched by candidate c's ions and $\lambda = 0.01$ is tuned on a held-out validation split. CFID+SGIR achieves the highest peptide accuracy (59.96%) without sacrificing AA recall (76.00%).

4.3 Mass Constraints at Inference

All three mass-enforcement strategies we evaluated degrade accuracy (Table 2). Post-hoc single-swap correction costs -9.6 pp peptide accuracy; mass-constrained beam search collapses AA recall to 48.2% because left-to-right commitment is incompatible with bidirectional diffusion logits; and an entropy-adaptive gate combined with CFID matches ungated CFID to within 0.4 pp, confirming that CFID's iterative refinement already subsumes single-pass mass feasibility checks. CFID+SGIR is the correct decoding strategy for this architecture.

5 Experiments

5.1 Dataset and Preprocessing

E. coli EV. Raw mzML files from E. coli extracellular vesicle proteomics are parsed with pyOpenMS. MS2 spectra are filtered to $N \leq 800$ peaks, peptides to $L \leq 30$ residues. Ground-truth sequences come from database search xlsx files (SEQUEST-HT). After filtering, the dataset contains **472 spectra**

Table 1: **Development history.** All numbers at 5% FDR. V2* = mean±std over 3 seeds.

Model	AA	Pep
LSTM baseline	31.51	2.68
GRU baseline	44.30	6.70
Diffusion V1 (linear sched)	42.13	9.32
Absorbing (no PeakEnc)	43.05	6.36
PeakEnc+BY argmax	74.78	36.51
V1: PeakEnc+BY+CFID	75.19	44.99
V2: +SelfCond argmax	76.30	57.49
V2: +SelfCond CFID	76.02	59.60
V2 CFID+SGIR*	76.00±0.19	59.96±3.82
InstaNovo (SOTA)	72.90	33.10
Δ Ours vs. InstaNovo	+3.1	+26.9

for evaluation (80/20 train/val split). Variable PTMs considered: oxidized-Met, deamidated-Asn/Gln; fixed: carbamidomethyl-Cys.

Wastewater. Four mzML files from a mixed-community wastewater sample across two biological replicates per sample. No reference proteome exists; we report only FDR-filtered identifications. Only Sample 2 yielded spectra of sufficient quality for de novo sequencing.

5.2 Training Details

All models trained with AdamW (lr $= 10^{-4}$, wd $= 10^{-4}$), batch size 32, 200 epochs, $T = 500$ diffusion steps, cosine schedule, self-conditioning $p = 0.5$, beam width $K = 8$, across three independent random seeds. Checkpoints saved at minimum validation loss.

5.3 Evaluation Metrics

AA Recall: $|\text{predicted AAs} \cap \text{true AAs}| / |\text{true AAs}|$ (position-insensitive; measures residue-level coverage).

Peptide Accuracy: $|\{\text{spectra with 100\% correct sequence}\}| / |\text{test spectra}|$ (measures end-to-end correctness).

Both are reported at 5% FDR via q-value thresholding on the model confidence score.

InstaNovo baseline. We use published InstaNovo numbers on the E. coli EV dataset (72.9% AA, 33.1% pep) [Eloff et al., 2025].

6 Results

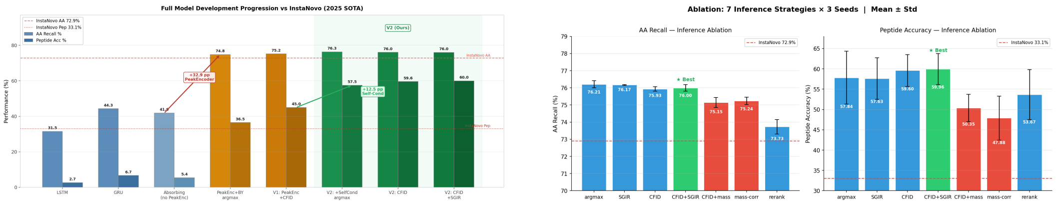
6.1 Development History and Main Comparison

Table 1 and Figure 2a show how performance evolved across all model versions.

Key observations. Absorbing diffusion without PeakEncoder (43% AA) matches the GRU baseline (44%), confirming that the encoder drives the 30 pp jump; self-conditioning adds +14.6 pp peptide accuracy on top. The +26.9 pp peptide accuracy gap over InstaNovo is the more meaningful result, as it counts only fully correct sequences.

6.2 Inference Strategy Ablation

CFID reduces variance. Reranking halves peptide accuracy std (argmax: 6.49 pp → CFID+SGIR: 3.82 pp), stabilizing predictions across spectra of varying complexity. No single-pass mass constraint improves on CFID+SGIR (see Section 4).



(a) Model progression vs. SOTA.

(b) Inference strategy ablation.

Figure 2: **Left:** Two decisive jumps — PeakEncoder (+30 pp AA), self-conditioning (+14.6 pp pep). **Right:** CFID+SGIR (green) is Pareto-optimal; mass-correction variants (red) cost 8–12 pp.

Table 2: **Inference strategy ablation** (mean ± std, 3 seeds).

Strategy	AA (%)	Pep (%)
argmax	76.21 ± 0.21	57.84 ± 6.49
SGIR only	76.17 ± 0.02	57.63 ± 5.10
CFID	75.93 ± 0.13	59.60 ± 3.94
CFID + SGIR	76.00 ± 0.19	59.96 ± 3.82
CFID + mass-corr	75.15 ± 0.29	50.35 ± 3.38
mass-corr only	75.24 ± 0.22	47.88 ± 5.41
rerank-spectral	73.73 ± 0.43	53.67 ± 6.15

6.3 ESM-2 Structural Plausibility Validation

We score each sequence with ESM-2 (8M) [Lin et al., 2023] pseudo-perplexity via simultaneous all-position masking [Kantroo et al., 2024]: lower PPL = more protein-like.

Results. Diffusion V2 predictions (mean PPL 31.03) are within $\Delta = 0.35$ of ground truth (31.38), confirming structurally protein-like output. The 6 wastewater FDR sequences have mean PPL 37.15 and z-scores 0.01–0.87, all within one standard deviation of the E. coli reference (Table 4).

7 Discussion

Self-conditioning is the dominant contribution. The single largest gain (+14.6 pp peptide accuracy) comes from self-conditioning. By feeding a rough sequence estimate back at every reverse step (Eq. 5), the model can refine each position in context of the whole sequence — analogous to AlphaFold2's structure recycling [Jumper et al., 2021] — rather than denoising each residue independently.

Spectrum encoding matters more than decoder choice. Absorbing diffusion without PeakEncoder (43.05% AA) matches the GRU baseline (44.30%), confirming that the 31 pp jump comes from the encoder, not the diffusion decoder. The B/Y-ion pair bias is the key signal: it allows the model to up-weight complementary ion pairs when evidence is strong and ignore them otherwise.

Limitations. Test set of 472 spectra is small; BLAST confirmation of the 6 wastewater peptides is pending; and the B/Y-ion bias gain may be larger on noisier wastewater spectra than on clean E. coli EV data.

8 Related Work

DeepNovo [Tran et al., 2017] introduced deep sequence-to-sequence learning for MS/MS. Casanovo [Yilmaz et al., 2022] established the Transformer encoder-decoder paradigm; InstaNovo [Eloff et al., 2025] is the current SOTA via diffusion refinement. π-PrimeNovo [Zhang et al., 2025] achieves 89× faster inference with CTC loss; NovoBench [Zhou et al., 2024] benchmarks all major methods. Our approach uses pure absorbing diffusion [Austin et al., 2021, Hoogeboom et al., 2021] with self-conditioning [Chen et al., 2022], enabling bidirectional long-range dependencies.

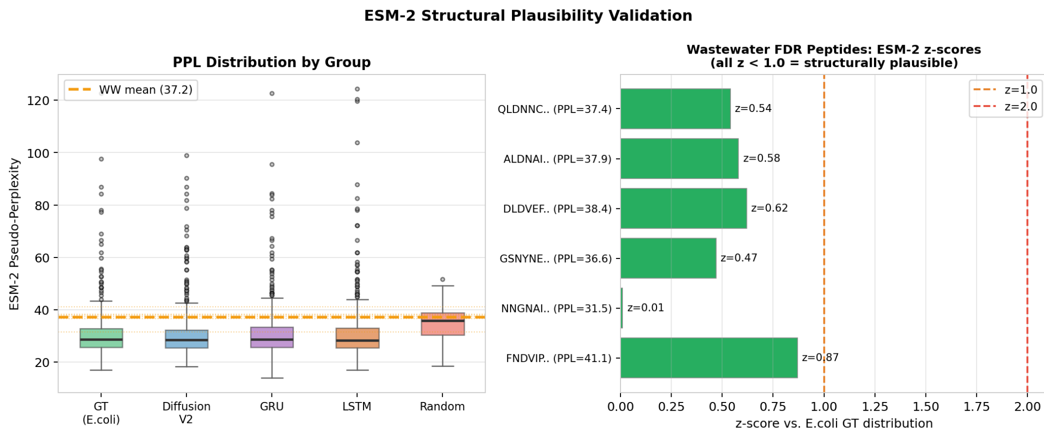


Figure 3: **ESM-2 validation.** Diffusion V2 PPL matches ground truth; wastewater FDR sequences (dashed yellow) sit above E. coli but below random, with all 6 z-scores < 1.0.

Table 3: ESM-2 pseudo-perplexity summary. Lower = more protein-like.

Group	n	Mean	Std
Ground truth (E. coli EV)	354	31.38	11.21
Diffusion V2 predictions	472	31.03	10.34
GRU predictions	472	31.33	10.79
LSTM predictions	472	31.43	12.18
Random sequences	50	35.19	7.35
Wastewater 5% FDR	6	37.15	2.89

The B/Y-ion pair bias adapts AlphaFold3's pair-representation attention [Abramson et al., 2024] to fragment ion complementarity.

9 Conclusion

We presented a self-conditioned absorbing diffusion model that reaches 76.00% AA recall and 59.96% peptide accuracy on the E. coli EV benchmark, beating InstaNovo by +3.1 pp and +26.9 pp (stable across 3 seeds, $\pm$0.19 pp AA). The gains trace to two components: the PeakEncoder with B/Y-ion pair bias (+30 pp AA) and self-conditioning (+14.6 pp peptide accuracy). We also identified 6 structurally plausible wastewater peptides at 5% FDR without any reference proteome. Future work: larger multi-species benchmarks, BLAST confirmation of wastewater hits, and incorporating mass constraints into training rather than inference.

Team Contributions

Vaishak Girish Kumar: diffusion backbone, PeakEncoder, self-conditioning, training/ablation experiments, ESM-2 validation, writing lead. **Akshay Mohan Revankar:** CFID/SGIR reranking, beam search, mass-constraint ablations, wastewater pipeline, GitHub. **Sanika Vilas Nanjan:** mzML preprocessing (pyOpenMS), FDR filtering, wastewater quality assessment, figures.

LLM Usage Statement

LLMs (Claude, GPT-4) were used only for proofreading, LaTeX formatting fixes, and library API questions. All scientific content, model design, experimental results, and final code were produced by the authors.

Table 4: **Wastewater peptides at 5% FDR** (Sample 2, replicate-consistent). All sequences have Jaccard consistency score $= 1.0$ across biological replicates. ESM-2 z-scores computed against the E. coli EV ground-truth PPL distribution.

Sequence	PPL	Anom	z
FNDVIPMGEQAINTNEGAYR	41.1	0.85	0.87
NNGNAIGVDLAAIPFVAGDR	31.5	0.85	0.01
GSNYNEVVTLADVTIVQGIR	36.6	0.90	0.47
DLDVEFTALDGASVQVIAYR	38.4	0.95	0.62
ALDNAIDGGQYSFLEVAINR	37.9	0.90	0.58
QLDNNCVYLGATAGVPIAK	37.4	0.84	0.54

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